

Luís Carlos Santos França

SOFTWARE ENGINEER

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INTRODUCTION

First-year Computer Engineering Masters student at the University of Minho, pursuing specializations in Applications Engineering and Knowledge Engineering. My primary interests lie in Databases and Data Engineering, with a strong inclination toward designing efficient and scalable data systems, where I can analyze data and extract meaningful insights. Currently looking for a summer internship to apply the skills I have acquired to real-world projects.

WORK EXPERIENCE

Volunteer Researcher

02/2026 - Ongoing

HASLab/INESC TEC — Braga, Portugal

- Engineered a **Deep Learning** based **Cardinality Estimator** based on Multi Set Convolutional Networks (MSCN), integrating it in an open source **Query Optimizer**.
- Developed a **PostgreSQL** logger to capture ground truth cardinality estimates and explore different possible execution sub plans.

Full-Stack Developer Intern

07/2025 - 08/2025

Mestreclique — Braga, Portugal

- Developed a full-stack web application using **ASP.NET Web Forms** and **C#**, focusing on dynamic page creation and robust **SQL Server** database integration (modeling, keys, and relationships).
- Responsible for developing and consuming **APIs/Web Services**, version control via **Git/GitHub**, and authoring technical documentation, ensuring scalability and adherence to development best practices.

Youth Robotics Instructor

06/2025 – 07/2025

Project "Programar o Futuro" — Braga, Portugal

- Instructed youths in robotics and programming logic through hands-on **Arduino** projects and sensor integration. Responsible for the pedagogical planning of the "**Programar o Futuro**" initiative, a project by SIC Esperança in collaboration with CCTIC-ESE/IPS and Google.org.

TECHNICAL SKILLS

Programming Languages	Java, Python, C#, TypeScript, Rust
Back-End	.NET, Spring Boot, Node.js
Databases	PostgreSQL, SQL Server, MongoDB, Snowflake, Qdrant, Neo4j
Front-End	Vue.js, Next.js, HTML5, TailwindCSS
Data & AI	Scikit-learn, PyTorch, Pandas, RAG, LLMs, Embedding
DevOps	Google Cloud, Docker, Kubernetes, Ansible, Git, Github Actions

EDUCATION

MSc in Computer Engineering

2025 – Present

University of Minho – Braga, Portugal

*Specializing in **Knowledge Engineering** and **Applications Engineering***

BSc in Computer Engineering

2022 – 2025

University of Minho – Braga, Portugal

SOFT SKILLS

- Team collaboration and communication
- Analytical thinking and problem solving
- Responsibility and continuous learning

LANGUAGES

- Portuguese — Native
- English — Proficient

FEATURED PROJECTS

Ekonomix - Investment Dashboard (Ongoing)

Project in Knowledge Engineering

- Finance **Dashboard** with stock, ETF and Crypto data
 - Data mining, and **ETL** pipeline scripts using **Prefect** for orchestration.
 - Orchestrated LLM workflows with **MLOps** to automate data source selection and query routing.
 - Implemented a multiple databases **MongoDB**, **Postgres** (with TimescaleDB extension), **Qdrant** optimized for time-series, relational, and vector data.
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Real-Time Bus Management System (Ongoing)

Project in Applications Engineering

- Full-stack system using **Spring Boot**, **Vue.js**, **PostgreSQL**, and **Redis**.
 - Real-time bus tracking using a Publisher-Subscriber system between drivers and clients using Redis, with QR code ticket validation.
 - Deployed on **Google Cloud Kubernetes**, with **load testing** and database benchmarking/optimization.
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Crypto Basket Predictor

Hackathon Bugsbyte

- Built a web platform to predict and visualize cryptocurrency basket prices using open-source ML models.
 - Full-stack development with **Vue.js**, **Django**, and **MySQL**.
 - Implemented **Time-Series Forecasting** models to predict cryptocurrency market trends.
 - Built a data pipeline to fetch and store historical market data for model training.
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PostgreSQL database optimization

Database Administration

- Analyzed both **transactional** and **analytical** queries to improve performance
 - Adjusted **SQL** code, created **Indexes**, changed **system parameters**, parallelism settings, etc.
 - **Benchmarking** in a Google Cloud virtual machine to ensure reproducibility of results.
 - Grade: 18/20
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Cloud Deployment of Sveltekit Webapp

Cloud Computing Applications and Services

- Automated cloud deployment using **Ansible** and **Kubernetes** (GKE) to a **GCP** environment.
 - Observed system state with Google Cloud Monitoring tools and used **Apache JMeter** for benchmarking.
 - Optimized performance and availability under load via auto-scaling and load balancing.
 - Grade 19,4/20
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Predicting road traffic flow

Data and Machine Learning

- Followed the **CRISP-DM** process of Data Mining to predict Porto's urban traffic flow, integrating external features like weather, school holidays, and sports events.
- **EDA** with **pandas** and **seaborn** libraries
- Applied **sklearn** and **pytorch** Machine Learning algorithms for predictions and analysis (Regressions, Ensemble, Clustering, Neural Networks, LSTM).
- Grade: 18,3/20